Cookie Puzzle

Aunt Geraldine knows that one of Alice, Bruno, Cecilia, Daniela or Eduardo ate all the cookies. She also knows that the guilty always lie and the innocent always tell the truth.

- Bruno says: "The culprit is either Eduardo or Cecilia."
- Eduardo says: "The guilty person is a girl."
- Finally, Daniela says: "If Bruno is guilty, then Cecilia is innocent."

Who ate the cookies, and why?

Predicate Logic Solution

To determine who ate the cookie, we can solve the problem using **predicate logic**. The solution follows all necessary steps, including defining constants and predicates, formulating the statements in logical form, and continuing with the transformation of these statements into CNF. The application of the **resolution method** is done the final conclusion.

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1. Vocabulary

Based on the problem statement, the main elements of the formal language are the constants: Bruno, Cecilia, and Eduardo.

Constants

Constants represent the specific individuals mentioned in the problem.

Object	Constant	Information
Ana	a	female
Bruno	b	male
Cecilia	c	female
Daniela	d	female
Eduardo	e	male

Variables

- x → a variable representing an arbitrary individual in the problem domain.
- y → another variable representing an arbitrary individual in the problem domain.

Predicates

- $girl(x)$ → a predicate that is true if and only if the individual x is girly.

2. Knowledge Base Formalization

1. At least one person is guilty

$$\exists X.guilty(X)$$

2. Exactly one person is guilty

$$\exists X.[guilty(X) \wedge \forall Y.(Y \neq X) \rightarrow \neg guilty(Y)]$$

3. Bruno says: "The culprit is either Eduardo or Cecilia".
$$guilty(e) \vee guilty(c)$$

4. Eduardo says: "The guilty person is a girl".
$$\forall X.guilty(X) \rightarrow [girl(x)]$$

5. Daniela says: "If Bruno is guilty, then Cecilia is innocent".
$$guilty(b) \rightarrow [\neg guilty(c) \vee \neg guilty(c)]$$

3. CNF Transformation

1. At least one is guilty
Formula

$$\exists X.guilty(X)$$

Conversion to CNF:

$$\exists X.guilty(X) \equiv guilty(x) \vee guilty(y) \vee guilty(z) \vee guilty(w) \vee guilty(v)$$

Resulting clauses

$$guilty(a) \vee guilty(b) \vee guilty(c) \vee guilty(d) \vee guilty(e)$$

2. Exactly one person is guilty
Formula

$$\exists X.[guilty(X) \wedge \forall Y.(Y \neq X) \rightarrow \neg guilty(Y)]$$

Conversion to CNF:

- Eliminate the quantifiers:
  - $X \neq Y \equiv [x, y, z, w, v]$
  - $\exists X.guilty(X) \equiv guilty(x) \vee guilty(y) \vee guilty(z) \vee guilty(w) \vee guilty(v)$ (at least one guilty)
  - $\forall Y.(Y \neq X) \rightarrow \neg guilty(Y) \equiv \neg guilty(x) \wedge \neg guilty(y) \wedge \neg guilty(z) \wedge \neg guilty(w) \wedge \neg guilty(v)$ (at least one guilty)
- Control clause:

$$\begin{aligned}
&(guilty(x) \vee guilty(y) \vee guilty(z) \vee guilty(w) \vee guilty(v)) \\
&\wedge (\neg guilty(x) \vee \neg guilty(y)) \\
&\wedge (\neg guilty(x) \vee \neg guilty(z)) \\
&\wedge (\neg guilty(x) \vee \neg guilty(w)) \\
&\wedge (\neg guilty(x) \vee \neg guilty(v)) \\
&\wedge (\neg guilty(y) \vee \neg guilty(z)) \\
&\wedge (\neg guilty(y) \vee \neg guilty(w)) \\
&\wedge (\neg guilty(y) \vee \neg guilty(v)) \\
&\wedge (\neg guilty(z) \vee \neg guilty(w)) \\
&\wedge (\neg guilty(z) \vee \neg guilty(v)) \\
&\wedge (\neg guilty(w) \vee \neg guilty(v))
\end{aligned}$$

Resulting clauses

$$\begin{aligned}
&guilty(a) \vee guilty(b) \vee guilty(c) \vee guilty(d) \vee guilty(e) \\
&\neg guilty(a) \vee \neg guilty(b) \\
&\neg guilty(a) \vee \neg guilty(c) \\
&\neg guilty(a) \vee \neg guilty(d) \\
&\neg guilty(a) \vee \neg guilty(e) \\
&\neg guilty(b) \vee \neg guilty(c) \\
&\neg guilty(b) \vee \neg guilty(d) \\
&\neg guilty(b) \vee \neg guilty(e) \\
&\neg guilty(c) \vee \neg guilty(d) \\
&\neg guilty(c) \vee \neg guilty(e) \\
&\neg guilty(d) \vee \neg guilty(e)
\end{aligned}$$

3. Bruno says: "The culprit is either Eduardo or Daniela".
Formula

$$guilty(e) \vee [guilty(c) \vee guilty(d)]$$

Conversion to CNF:

$$\neg guilty(e) \vee [guilty(c) \vee guilty(d)]$$

$$\begin{aligned}
&\neg guilty(e) \vee [guilty(c) \vee guilty(d)] \equiv (\neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(a) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(b) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(c) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(d) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(d) \vee \neg guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(e) \vee \neg guilty(c) \vee guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(c) \vee \neg guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(d) \vee \neg guilty(c) \vee \neg guilty(d)) \\
&\wedge (\neg guilty(e) \vee \neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(d))
\end{aligned}$$

Resulting clauses

$$\begin{aligned}
&\neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(a) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(b) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(c) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(d) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(e) \vee \neg guilty(e) \vee guilty(c) \vee guilty(d) \\
&\neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(c) \vee guilty(d) \\
&\neg guilty(e) \vee \neg guilty(d) \vee \neg guilty(c) \vee guilty(d) \\
&\neg guilty(e) \vee \neg guilty(e) \vee \neg guilty(c) \vee guilty(d) \\
&\neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(c) \vee \neg guilty(d) \\
&\neg guilty(e) \vee \neg guilty(d) \vee \neg guilty(c) \vee \neg guilty(d) \\
&\neg guilty(e) \vee \neg guilty(e) \vee \neg guilty(c) \vee \neg guilty(d)
\end{aligned}$$

4. Eduardo says: "The guilty person is a girl".
Formula

$$\forall X.guilty(X) \rightarrow [girl(x)]$$

Conversion to CNF:

- Eliminate implications:

$$guilty(X) \rightarrow [girl(x)] \equiv \neg guilty(X) \vee [girl(x)]$$

- Expand equivalences:

$$\begin{aligned}
&\neg guilty(X) \vee [girl(x)] \equiv (\neg guilty(X) \vee girl(x)) \\
&\wedge (\neg guilty(a) \vee girl(x)) \wedge (\neg guilty(b) \vee girl(x)) \\
&\wedge (\neg guilty(c) \vee girl(x)) \wedge (\neg guilty(d) \vee girl(x)) \\
&\wedge (\neg guilty(e) \vee girl(x)) \wedge (\neg guilty(v) \vee girl(x))
\end{aligned}$$

Resulting clauses

$$\begin{aligned}
&\neg guilty(X) \vee girl(x) \\
&\neg guilty(a) \vee girl(x) \\
&\neg guilty(b) \vee girl(x) \\
&\neg guilty(c) \vee girl(x) \\
&\neg guilty(d) \vee girl(x) \\
&\neg guilty(e) \vee girl(x) \\
&\neg guilty(v) \vee girl(x)
\end{aligned}$$

5. Daniela says: "If Bruno is guilty, then Cecilia is innocent".
Formula

$$guilty(b) \rightarrow [\neg guilty(c) \vee \neg guilty(c)]$$

Conversion to CNF:

- Eliminate implications:

$$guilty(b) \rightarrow [\neg guilty(c) \vee \neg guilty(c)] \equiv \neg guilty(b) \vee [\neg guilty(c) \vee \neg guilty(c)]$$

- Expand equivalences:

$$\begin{aligned}
&\neg guilty(b) \vee [\neg guilty(c) \vee \neg guilty(c)] \equiv (\neg guilty(b) \vee \neg guilty(c)) \\
&\wedge (\neg guilty(b) \vee \neg guilty(c)) \equiv (\neg guilty(b) \vee \neg guilty(c))
\end{aligned}$$

Resulting clauses

$$\begin{aligned}
&\neg guilty(b) \vee \neg guilty(c) \\
&\neg guilty(a) \vee \neg guilty(c) \\
&\neg guilty(b) \vee \neg guilty(c) \\
&\neg guilty(c) \vee \neg guilty(c) \\
&\neg guilty(d) \vee \neg guilty(c) \\
&\neg guilty(e) \vee \neg guilty(c) \\
&\neg guilty(c) \vee \neg guilty(c)
\end{aligned}$$

4. Knowledge Base Clauses

After converting the formal representation of the knowledge base into CNF, we removed all duplicates and observed the final collection of clauses:

No	Knowledge Base Clause
1	$guilty(a) \vee guilty(b) \vee guilty(c) \vee guilty(d) \vee guilty(e)$
2	$\neg guilty(a) \vee \neg guilty(b)$
3	$\neg guilty(a) \vee \neg guilty(c)$
4	$\neg guilty(a) \vee \neg guilty(d)$
5	$\neg guilty(a) \vee \neg guilty(e)$
6	$\neg guilty(b) \vee \neg guilty(c)$
7	$\neg guilty(b) \vee \neg guilty(d)$
8	$\neg guilty(b) \vee \neg guilty(e)$
9	$\neg guilty(c) \vee \neg guilty(d)$
10	$\neg guilty(c) \vee \neg guilty(e)$
11	$\neg guilty(d) \vee \neg guilty(e)$
12	$\neg guilty(e) \vee girl(x)$
13	$\neg guilty(a) \vee girl(x)$
14	$\neg guilty(b) \vee girl(x)$
15	$\neg guilty(c) \vee girl(x)$
16	$\neg guilty(d) \vee girl(x)$
17	$\neg guilty(v) \vee girl(x)$

5. Resolution with Set-of-Support Strategy

The **set-of-support resolution** strategy is a logical refinement strategy in which resolution is applied only to clauses that involve a specific subset of all clauses (typically related to the goal being proved). The approach restricts the search space, focusing on clauses directly relevant to the goal and helps avoid irrelevant combinations in the derived conclusion more efficiently.

Case Alex

- All clauses are divided into two sets:
  - **Knowledge Base (KB)** → contain facts and rules
  - **Set of Support (SOS)** → clauses related to the negation of the goal

Resolution rule

- Resolution is applied **only if at least one parent clause belongs to the SOS**.

Why it works

- Reduces the search space
- Prevents unnecessary resolutions
- Guides reasoning toward a contradiction or proof

Goal

- Derive the **empty clause** ($\square$), which confirms the conclusion.

Results

- More efficient and structured logical inference

5.1 Daniela

First, we test the hypothesis that **Daniela is innocent**: $innocent(d)$. To do this, we apply the resolution method in order to confirm or refute this assumption.

No	Derived Clause	Resolution Step / Description
17	$innocent(d)$	Negated conclusion
18	$\neg guilty(b)$	Resolution of clauses 16 and 17
19	$guilty(c)$	Resolution of clauses 16 and 17
20	$\neg guilty(a)$	Resolution of clauses 6 and 18
21	$\square$	Resolution of clauses 19 and 20

As a result of the resolution process, an **empty clause** is derived, which indicates a logical contradiction when the negation of the hypothesis is assumed. Therefore, the assumption $innocent(d)$ is true, and we can state to the knowledge base: **Ana is innocent**.

5.2 Cecilia

Knowing that **Daniela is innocent**, we can assume that her statement is true: **If Bruno is guilty, then Cecilia is innocent**. Based on this, we test the hypothesis that Cecilia is innocent: $innocent(c)$.

No	Derived Clause	Resolution Step / Description
17	$innocent(c)$	Negated conclusion
18	$\neg guilty(b)$	Resolution of clauses 6 and 18
19	$guilty(a)$	Resolution of clauses 6 and 18
20	$\neg guilty(a)$	Resolution of clauses 18 and 19
21	$\square$	Resolution of clauses 19 and 20

As a result of the resolution process, an **empty clause** is derived, which indicates a logical contradiction when the negation of the hypothesis is assumed. Therefore, the assumption $innocent(c)$ is false, and we can add to the knowledge base: **Cecilia is guilty**.

5.3 Bruno

Now, in order to fully verify Daniela's statement, we attempt to prove the hypothesis that **Bruno is guilty**: $guilty(b)$.

No	Derived Clause	Resolution Step / Description
22	$\neg guilty(b)$	Negated conclusion
23	$\neg guilty(c) \vee girl(x)$	Resolution of clauses 10 and 19
24	$\neg guilty(c)$	Resolution of clauses 10 and 20
25	$\neg guilty(c)$	Resolution of clauses 9 and 20
26	$\neg guilty(d)$	Resolution of clauses 10 and 24
27	$\square$	Resolution of clauses 20 and 26

During the resolution process, **no empty clause was derived**. Instead, the following literals were obtained in the course of resolution: $\neg guilty(c) \vee girl(x)$, $\neg guilty(c)$, $\neg guilty(d)$.

This means that we **cannot prove** that Bruno is guilty. Therefore, we proceed to test the alternative hypothesis — that **Bruno is innocent**.

No	Derived Clause	Resolution Step / Description
22	$innocent(b)$	Negated conclusion
23	$guilty(c) \vee girl(x)$	Resolution of clauses 8 and 19
24	$\neg guilty(c)$	Resolution of clauses 8 and 19
25	$guilty(d) \vee girl(x)$	Resolution of clauses 8 and 19
26	$girl(x) \vee girl(x)$	Resolution of clauses 20 and 23
27	$girl(x)$	Resolution of clauses 20 and 24
28	$\square$	Resolution of clauses 24 and 27

As a result of the resolution with the negation of the hypothesis that Bruno is guilty, an **empty clause** was obtained, which proves the contrary. Consequently, we add the fact $innocent(b)$ to the knowledge base as the **numbered 19**.

5.4 Eduardo

We already have proofs that **Cecilia** and **Bruno** are innocent. Therefore, Daniela's statement has been fully analyzed, and we can proceed to the next statement — **Bruno's**.

Bruno claims that the guilty person is either **Eduardo** or **Daniela**. However, since we already know that **Daniela is innocent**, we can test the hypothesis that **Eduardo is guilty**: $guilty(e)$.

No	Derived Clause	Resolution Step / Description
20	$\neg guilty(e)$	Negated conclusion
21	$guilty(c) \vee girl(x)$	Resolution of clauses 12 and 20
22	$\neg guilty(c)$	Resolution of clauses 10 and 20
23	$\square$	Resolution of clauses 19 and 22

As a result of the resolution process, we obtained an **empty clause**, which means that the hypothesis of Eduardo's guilt is true. Thus, we have identified the person who ate the cookies.

5.5 Ana

Since the guilty person — **Eduardo** — has been identified, there is no need to continue checking further hypotheses, in particular whether **Ana** is guilty or not. The statement that one of the girls ate the cookie is false, therefore **Ana is innocent**. Consequently, **Cecilia** is the only remaining girl and an additional resolution proof.

6. Automated Solver Verification

To ensure the correctness of our solution, we will verify it using automated SAT/SMT solvers, namely **DPLL SAT Solver** and **CD Playground**.

6.1 DPLL SAT Solver

DPLL SAT Solver is an implementation of the **Davis-Putnam-Logemann-Loveland (DPLL)** algorithm for checking the satisfiability of propositional logic formulas in conjunctive normal form (CNF). This algorithm uses **unit clause propagation** and **clause simplification** to efficiently explore search spaces.

To run the DPLL solver correctly, our knowledge base is transformed into the format it accepts:

- The $\neg$ (NOT) symbol is represented, leaving only the literals represented as CNF clauses.
- A minus (-) indicates the negation of a literal.

Example Input:

```
-a b c d e
-a -b
-a -c
-a -d
-a -e
-b -c
-b -d
-b -e
-c -d
-c -e
-d -e
e c d
-e c
-e c d
-c -c
-d c
-e c
-c -c
```

Solution:

```
satisfiable s -a -b -d -c
```

6.2 CD Playground

CD Playground is a web interface for the **CD SAT solver**, which allows checking the satisfiability of logical formulas written in various notations, including TPTP, Supported Boolean, Integer, and other types of variables, enabling the formalization of more complex logical relationships.

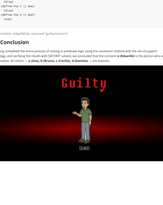
For CD, our knowledge base is translated into the correct TMT-LIB format.

```
; ==== Variables ====
(declare-fun a () Bool) ; ana
(declare-fun b () Bool) ; bruno
(declare-fun c () Bool) ; cecilia
(declare-fun d () Bool) ; daniela
(declare-fun e () Bool) ; eduardo

; ==== Exactly one guilty ====

; At least one
(assert (or a b c d e))

; Exactly one
(assert (or (and a (not b) (not c) (not d) (not e))
            (and (not a) b (not c) (not d) (not e))
            (and (not a) (not b) c (not d) (not e))
            (and (not a) (not b) (not c) d (not e))
            (and (not a) (not b) (not c) (not d) e))

; ==== Bruno ====
; Bruno says: "The guilty person is a girl" = e or c or d
(assert (or (not b) (or e c d)))

; ==== Eduardo ====
; Eduardo says: "The guilty person is a girl" = e or c or d
(assert (or (not e) (or c d)))

; ==== Daniela ====
; Daniela says: "If Bruno is guilty, then Cecilia is innocent"
(assert (or (not b) (and (not c) (not c))))

; ==== Check ====
(check-sat)
(get-model)
```

Output:

```
sat
(
  (a
    (declare-fun a () Bool
    false)
  (b
    (declare-fun b () Bool
    false)
  (c
    (declare-fun c () Bool
    false)
  (d
    (declare-fun d () Bool
    false)
  (e
    (declare-fun e () Bool
    true)
)
```

Variables: b=false,c=false,d=false,e=true

7. Conclusion

Having completed the entire process of solving a predicate logic using the resolution method with the set-of-support strategy, we can now confirm that the person who ate the cookies was **Eduardo**. Although **Bruno** and **Eduardo** initially claimed that they were the culprits, the solution — **Eduardo** — was **Bruno's**, **Cecilia's**, **Daniela's**, and **Ana's** favorite.